

MMGrip: A Handheld Multimodal Haptic Device Combining Vibration, Impact, and Shear for Realistic Expression of Contact

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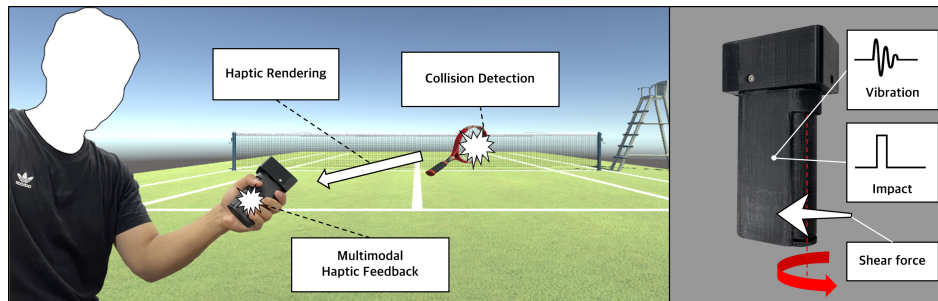


Figure 1: MMGrip is a handheld haptic device that combines three types of tactile stimulus—vibration, impact, and shear—to improve the realism of haptic stimuli representing virtual collisions.

ABSTRACT

We introduce MMGrip, a handheld multimodal haptic device that simultaneously presents vibration, impact, and shear for realistic and immersive haptic feedback to virtual collision events. The three types of haptic stimulus mimic a damped vibration, a collision impulse, and a skin slip deformation, respectively, occurring at physical contact. Controlling the three stimuli, as well as their onset time differences, provides a vast design space for offering diverse and realistic haptic experiences. We present the design of MMGrip and demonstrates its application to virtual sports.

CCS CONCEPTS

• Human-centered computing → Haptic devices; • Hardware → Haptic devices.

KEYWORDS

Haptics, Virtual reality, Multimodal, Handheld

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1 INTRODUCTION

We present MMGrip, a handheld haptic device aiming to provide realistic and rich haptic experiences during interaction in virtual reality (VR) applications. Our work was motivated by two critical current limitations of the related research and technology. First, current commercial VR headsets generally employ handheld controllers for interaction, but their haptic feedback function remains elementary using only vibration actuators. Researchers demonstrated handheld devices with advanced haptic mechanisms, e.g., [Benko et al. 2016; Choi et al. 2018], but such principles may not be very practical for handheld devices. Second, we see increased attention on *multimodal* haptics, where a few different types of haptic actuators simultaneously trigger multiple haptic stimuli to diversify the range of haptic expressions [Wang et al. 2019]. Simply adding vibration and impact stimuli can considerably improve the realism of virtual object collisions [Park et al. 2019].

MMGrip is a project in which we have been designing and testing a *pragmatic* handheld haptic device for VR by utilizing three types of haptic mode: vibration, impact, and shear. Our emphasis is on using only off-the-shelf components in a simple structural packaging while maximizing the users' benefits of perceptual experiences, for easy reproduction. We demonstrate an application of MMGrip to virtual sports by expressing virtual collisions; see Figure 1. This multimodal haptic device affords a testbed for crafting effective haptic effects by rigorous perceptual validation, on which we are focusing now.

2 MMGRIP

2.1 Mechanical Design

Our goal is to deliver all three types of haptic stimulus to the user's palm grasping MMGrip. To this end, our prototype compactly packages a vibration actuator, an impact actuator, and a cylinder for

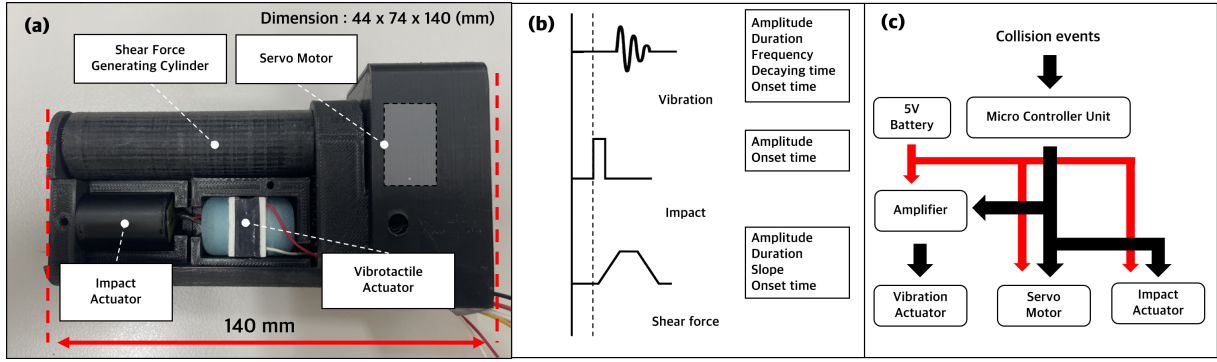


Figure 2: (a) MMGrip prototype showing hardware components. (b) Profiles of multimodal haptic stimuli. (c) Diagram of hardware components.

shear feedback all within the device handle, as shown in Figure 2(a). Other electronic components, such as an MCU, an amplifier, and a battery, are assembled in a box at the top, which also provides reasonable sensations of weight and wielding for dynamic touch perception [Park et al. 2021]. The dimension of the prototype is $44 \times 74 \times 140$ mm, and the weight is 170 g.

A strong contact impulse is generated initially at collision with a physical object. For its expression, we adopt a strong impact actuator (DHA2030, Acturionix, $\varnothing 20 \times 30$ mm). To generate damped transient vibratory responses after impact, we use a wideband voice-coil actuator (Haptuator BM2C, Tactile Labs, $37.5 \times 20 \times 20$ mm). Such a combination can improve the realism of contact simulation to the great extent [Park et al. 2019]. The maximum output is 30 G for impact and 2 G at 100 Hz for vibration. Lastly, the skin deformation during collision is emulated by presenting a shear force to the palm. The direction of a short tactile stimulus, either perpendicular or tangential to the skin, is indistinguishable to users. Using a servo motor (MG92B, Towerpro, $22.8 \times 12 \times 31$ mm), we substitute the normal skin indentation with a lateral skin stretch. Shear feedback alone can create convincing illusions of haptic events [Tactical Haptics 2022].

The diagram of the hardware system is shown in Figure 2(c). Our hardware system is powered by a 5V battery, and one MCU (ESP32) generates signals for all multimodal haptic stimuli.

2.2 Stimulus Design Space

Design of multimodal haptic stimuli for MMGrip requires to determine many variables; see Figure 2(b). At present, we model a vibration stimulus using an exponentially decaying sinusoidal function, which is widely used to represent collision events. This model has four parameters: frequency, amplitude, decaying rate, and duration. For an impulse stimulus, we control only its amplitude. The impulse actuator we use does not allow duration modulation. A shear force is commanded using the conventional trapezoidal profile. It has three parameters of amplitude, duration, and slope (same for the increasing and decreasing intervals). Another critical parameters are the onset times of the three stimuli. They need to be carefully tuned while considering the target physical events and the actuator latencies. These 11 variables constitute the design space of haptic stimuli for MMGrip.

3 USE SCENARIOS

MMGrip can render various haptic events occurring during normal and lateral explorations of virtual objects. An example application is a VR racket game, where the user hits the ball using a virtual racket and feels a multimodal haptic collision response generated by MMGrip; see Figure 1. The pose of MMGrip is tracked by Azure Kinect and mapped into the virtual environment. According to pilot tests, MMGrip can provide compelling haptic experiences of collision by stimulating the user's palm with simultaneous impact, vibration, and shear stimuli.

4 CONCLUSIONS AND ONGOING WORK

MMGrip is a handheld multimodal haptic device that generates three types of haptic stimulus—impact, vibration, and shear—using only off-the-shelf components. MMGrip offers an exceptionally many design variables, and we are currently investigating their perceptual effects, both independent and correlated, to provide perceptually-optimized haptic effects for improved VR experiences. We are also eager to devise general haptic rendering algorithms that maximize the potential of the three multimodal stimuli for VR.

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